

Troubleshooting Guide for AI Code Generation Platforms

This guide helps users of AI-driven code generation platforms (e.g., Firebase Studio, Bolt, StudAI Builder) resolve common issues quickly.

1. Connectivity Issues

- **Symptoms:** Error messages like "NetworkError", "API unreachable"
- **Solutions:**
 1. Verify your internet connection
 2. Check API endpoint URL in settings
 3. Test endpoint: `curl -I https://api.yourplatform.com`
 4. Ensure API keys are valid and not expired

2. Build & Compilation Errors

- **Symptoms:** Missing modules, compilation failures
- **Solutions:**
 1. Match Node.js version with project (`node --version`)
 2. Clean install: `rm -rf node_modules && npm ci`
 3. Reinstall missing peer dependencies
 4. Clear caches: `npm cache clean --force`

3. Deployment Failures

- **Symptoms:** Build errors on deployment platforms
- **Solutions:**
 1. Confirm build output directory (`dist/,build/`)
 2. Verify environment variables set in deployment config
 3. Test build locally (`npm run build`)
 4. Review deployment logs for specific errors

4. Authentication Issues

- **Symptoms:** Login failures, 401 errors
- **Solutions:**
 1. Check authentication provider status (Firebase, Supabase)
 2. Validate auth configurations and redirect URLs

3. Inspect CORS settings for allowed origins

5. Database Connectivity

- **Symptoms:** Connection timeouts, permission errors
- **Solutions:**
 1. Verify connection string and credentials
 2. Check database firewall/IP whitelist
 3. Ensure SSL or TLS settings match requirements

6. Code Generation Errors

- **Symptoms:** Empty output, token limit errors
- **Solutions:**
 1. Improve prompt clarity with specific requirements
 2. Monitor API usage and quotas in dashboard
 3. Implement retry logic with exponential backoff

7. Styling & UI Issues

- **Symptoms:** Broken layouts, missing styles
- **Solutions:**
 1. Confirm theme or CSS imports are correct
 2. Check for conflicting CSS frameworks
 3. Validate design tokens or CSS variables

8. Performance Bottlenecks

- **Symptoms:** Slow load times, high CPU/memory usage
- **Solutions:**
 1. Use code splitting and lazy loading for heavy components
 2. Memoize expensive computations (`useMemo`, `React.memo`)
 3. Use performance monitoring tools (Lighthouse, DevTools)

Quick Debug Workflow

1. **Reproduce:** Identify steps to recreate issue consistently
2. **Inspect:** Check console/network logs
3. **Isolate:** Comment out or disable features to narrow scope
4. **Resolve:** Apply targeted fixes based on error context
5. **Verify:** Test on different environments (dev, staging)

Estimated reading time: 5 minutes